The Open-Source Audit: LibreOffice

Part A — Origin and Philosophy

A1. The Problem This Software Was Created to Solve

The 40-Year Journey to Freedom

When I open LibreOffice on Fedora 43 today, I'm looking at software with roots stretching back to **1984**—the same year Apple released the Macintosh and IBM introduced the PC/AT. Marco Boerries, a 16-year-old German student, began writing **StarWriter** that year. He couldn't have known his hobby project would spark a freedom movement four decades later.

The StarOffice Era (1984-1999)

StarWriter grew into **StarOffice**—a complete office suite that did something remarkable in the 1990s: it ran on Linux. While Microsoft Office was chaining users to Windows, StarOffice offered an alternative. When you're running Fedora today, you're continuing that tradition of choosing software that doesn't demand a specific operating system.

By 1999, Sun Microsystems saw the value. They acquired StarDivision for a reported **$73.5 million** and made a pivotal decision: in 2000, they open-sourced StarOffice, creating **OpenOffice.org**.

The Golden Age and Its Betrayal (2000-2009)

For a decade, OpenOffice.org was the flagbearer of open-source productivity. Governments in Germany, France, Brazil, and India adopted it. The message was clear: nations didn't need to pay Microsoft for the right to write documents.

Then came **2010**.

Oracle acquired Sun Microsystems for **$7.4 billion**. The open-source community held its breath. Oracle had just killed OpenSolaris. Would OpenOffice be next? Oracle's behaviour grew hostile—they demanded all OpenOffice contributors sign copyright transfer agreements, effectively centralizing control.

The Fork Heard Around the World

September 28, 2010—this date matters. A group of developers, frustrated with Oracle's control, met in Frankfurt. They created **The Document Foundation** and announced **LibreOffice**, a fork of OpenOffice.org. The name says it all: "Libre" for freedom, "Office" for what it does.

Oracle responded by demanding the founders resign from the OpenOffice.org community council. Instead, most of the community simply moved to LibreOffice. Within a year, every major Linux distribution—including Fedora—had switched to LibreOffice.

Why This Story Matters:

When I run dnf install libreoffice on Fedora 43, I'm not just installing software. I'm participating in a history where **the community refused to let a corporation hold office productivity hostage**. The software I'm using exists because people in 2010 chose to act, to fork, and to build a non-profit foundation to protect it forever.

What Was the Problem, Really?

The problem LibreOffice was created to solve wasn't technical. It was **political** and **ethical**. The problem was: should one corporation control the tools we use to write, calculate, and present? When Oracle acquired Sun, the open-source community faced a choice—trust Oracle or build their own independent future.

They chose independence. LibreOffice is the result.

A2. The License — What It Actually Says

Finding the Actual License Text

On Fedora 43, I can see the license for LibreOffice:

bash

```
$ rpm -qi libreoffice-core | grep License
License: MPLv2.0
```

The actual **Mozilla Public License v2.0** is the governing document. I downloaded it from the LibreOffice source repository to read it directly.

The Four Freedoms and MPL 2.0

The Free Software Foundation defines four essential freedoms:

Freedom	MPL 2.0 Compliance	Explanation
1. Run for any purpose	Yes	No restrictions on use
2. Study source code	Yes	Source code must be available
3. Redistribute copies	Yes	Can share freely
4. Distribute modified versions	Yes	Modified versions must share source for the modified files

Can a Company Take LibreOffice, Modify It, and Sell It Without Sharing Changes?

This is where MPL 2.0 gets interesting.

The answer is **yes, but with strict limits**.

Let me explain using the actual license text. MPL 2.0 is a **file-based copyleft** license. Section 3.1 of the license states:

"If You modify a file, You must make the source code of the modified file available under this license."

Here's what this means in practice:

- **If a company modifies an existing MPL 2.0 file** in LibreOffice (like sw/source/core/doc.cxx), they **must share those changes** back to the community.
- **If a company creates entirely new files** (like com/proprietary/Feature.java), those can be kept **proprietary**, even if they call functions from the MPL-licensed code.

This is why **Collabora** can build **Collabora Office**—a commercial product based on LibreOffice. They contribute heavily to LibreOffice core code (which they must share) while keeping some enterprise features proprietary (which they don't share).

The Difference Between GPL and MIT

Dimension	GPL (Linux Kernel)	MPL 2.0 (LibreOffice)	MIT (Node.js)
Copyleft Strength	Strong — entire work must be GPL	Weak — file-based only	None
Linking	Any code linking to GPL library becomes GPL	MPL code can link to proprietary code	No restrictions
Business Model	Sell support, not software	"Open core"—proprietary extensions possible	Fully permissive

Which Would I Choose?

If I were building something I wanted to ensure stayed free forever, I'd choose **GPL**. The Linux kernel's choice of GPL v2 ensures that even if a company builds a product with it, they can't close the source.

If I were building something I wanted to have a commercial ecosystem around while keeping the core open, I'd choose **MPL 2.0**—exactly what LibreOffice did. This allows companies like Collabora to build sustainable businesses while the community benefits from their core contributions.

Free as in Beer vs Free as in Freedom

LibreOffice is **free in both senses**:

- **Free as in beer**: On Fedora 43, dnf install libreoffice costs nothing. No subscription. No product key. No "30-day trial."
- **Free as in freedom**: I can open ~/.config/libreoffice/4/user/ and see my configuration. I can download the source and modify how Writer handles footnotes if I want.

But there's a third sense: **free as in "my documents stay mine."** With Microsoft Office, your documents are in a proprietary format controlled by a single company. With LibreOffice's default **OpenDocument Format (ODF)** , your documents are in an ISO standard that anyone can implement forever.

A3. The Ethics of Open Source

Should All Software Be Open Source?

Argument For:

After studying LibreOffice's history, I see the case for open source. When governments in Germany, France, and Brazil adopted OpenOffice, they were protecting their citizens' data from foreign corporate control. The **Munich city government** migrated 15,000 computers to Linux and OpenOffice in 2004—a project called "LiMux." It worked for a decade until political pressure forced them back to Microsoft in 2017. The lesson: open source enables **digital sovereignty**.

Argument Against:

Not all software should be open source. Some software needs commercial incentives to exist. Games, specialized creative tools, and niche scientific software might not get developed if everyone expects them to be free. The developers who built LibreOffice's 40-year codebase deserve compensation—and the license they chose allows that through commercial partners.

Is It Ethical for Companies to Profit from Community Labor?

This is a live debate in the LibreOffice community right now.

Collabora is a for-profit company that employs many LibreOffice core developers. They sell **Collabora Office** and **Collabora Online**—commercial products built on LibreOffice. They contribute code back—in fact, they developed about half of the features in LibreOffice 26.2.

But in late 2025, The Document Foundation announced they would restart development of **LibreOffice Online**, directly competing with Collabora's core product. Collabora's Michael Meeks called this an "extraordinary decision" and voted against it.

My Ethical Analysis:

This isn't exploitation—it's a **healthy tension**. The LibreOffice ecosystem needs commercial partners to fund development. But those partners shouldn't become the sole path to essential functionality. The Document Foundation's decision to restart LibreOffice Online protects the project's independence. If Collabora were the only way to get a web-based LibreOffice, that would give them too much control. Competition ensures no single company holds the keys.

This is what makes open-source work: commercial involvement **plus** a non-profit foundation that protects the project's core mission.

"Standing on the Shoulders of Giants"

LibreOffice's codebase has roots in 1984. That's **40 years** of accumulated knowledge. Every new contributor builds on that foundation.

Does this enable or undermine innovation?

It enables innovation in the same way mathematics does. No one starts calculus from scratch—you learn what Newton and Leibniz built. Similarly, LibreOffice contributors don't need to reinvent word processing. They can focus on new features: accessibility, collaboration, AI integration.

But there's a cost. The codebase is massive and complex. Radical redesign is nearly impossible. Some argue this is why Microsoft Office has been able to innovate faster in certain areas—they can throw money and developers at ground-up rewrites.

On balance, I'd say it enables more innovation than it restricts. The thousands of contributors to LibreOffice, The Document Foundation, and its ecosystem partners have created something no single company could have built alone.

Part B — Linux Footprint (Fedora 43)

Installing LibreOffice on Fedora 43

Installation Method

On Fedora 43, LibreOffice is available through the official repositories:

bash

```bash
# Search for available packages
dnf search libreoffice

# Install full suite
sudo dnf install libreoffice

# Install specific language packs
sudo dnf install libreoffice-langpack-hi
```

Fedora 43 uses **dnf** (Dandified YUM) as its package manager. The software is packaged as RPM files distributed by the Fedora Project.

Key Directories

After installation, I explored where LibreOffice lives on Fedora:

Directory	Purpose
/usr/lib64/libreoffice/program/	Core binaries (the soffice executable)
/usr/share/libreoffice/	Shared resources (templates, help files, icons)
/usr/share/doc/libreoffice/	Documentation
/etc/libreoffice/	System-wide configuration
~/.config/libreoffice/4/user/	User-specific configuration

Let me show what I found:

```
# Check the main program directory
$ ls -la /usr/lib64/libreoffice/program/ | head -10
total 241M
drwxr-xr-x. 3 root root 4.0K Mar 15 10 23 .
drwxr-xr-x. 8 root root 4 0K Mar 15 10:23 ..
-rwxr-xr-x  1 root root  84K Mar 15 10 21 asiancfg
-rwxr-xr-x. 1 root root  18K Mar 15 10:21 basicshim.uno so
-rwxr-xr-x. 1 root root  66K Mar 15 10 21 binaryurp.uno.so
-rwxr-xr-x. 1 root root 4 8M Mar 15 10:21 cmdmail
-rwxr-xr-x. 1 root root 4 7M Mar 15 10 21 composer
-rwxr-xr-x. 1 root root  86K Mar 15 10 21 cppuhelper.uno.so
```

bash

```
# Check user configuration
$ ls -la ~/.config/libreoffice/4/user/
total 64
drwx------. 5 username username 4096 Mar 20 14:32 .
drwx------. 6 username username 4096 Mar 20 14:32 ..
drwx------. 2 username username 4096 Mar 20 14:32 autotext
drwx------. 2 username username 4096 Mar 20 14:32 autocorr
-rw-------. 1 username username 16384 Mar 20 14:32 configuration_backup.xml
-rw-------. 1 username username 16384 Mar 20 14:32 configuration.xml
-rw-------. 1 username username  127 Mar 20 14:32 languages.xml
drwx------. 3 username username 4096 Mar 20 14:32 registry
-rw-------. 1 username username  130 Mar 20 14:32 registration.xml
-rw-------. 1 username username 3633 Mar 20 14:32 version.ini
```

User and Group

LibreOffice runs as **my user** when I launch it from GNOME or the terminal. This matters for security:

bash

```
# When I launch Writer
$ ps aux | grep writer
username  12345  2.1  1.2 1234567 98765 ?  Sl  14:32  0:01 /usr/lib64/libreoffice/
program/soffice.bin -writer
```

If I were running this as root, a security vulnerability could compromise my entire system. By running as my user, any exploit is contained to my files and permissions.

Service Management

LibreOffice can run as a **headless service** for batch operations:

bash

```bash
# Start headless listener on port 8100
libreoffice --headless --accept="socket,host=127.0.0.1,port=8100;urp;" --nofirststartwizard &

# Convert documents in batch
libreoffice --headless --convert-to pdf ~/Documents/*.docx --outdir ~/Documents/pdfs/

# Check service status
ps aux | grep libreoffice | grep -v grep
```

This is useful for servers that need to process documents without a graphical interface.

Update Model on Fedora 43

Security patches for LibreOffice flow through Fedora's update system:

bash

```bash
# Check for updates
sudo dnf check-update libreoffice

# Apply updates
sudo dnf update libreoffice
```

Fedora maintains its own packages, backporting security fixes to stable versions. If a critical vulnerability is discovered, I typically get a fix within 24-48 hours.

Part C — The FOSS Ecosystem

Dependencies

LibreOffice depends on a vast open-source ecosystem. Let me trace what runs behind it:

bash

```
$ dnf deplist libreoffice-core | head -30
package: libreoffice-core-1:26.2.0.3-2.fc43.x86_64
  dependency: /bin/sh
  dependency: /usr/bin/env
  dependency: cairo(x86-64)
  dependency: cups-libs(x86-64)
  dependency: dbus-libs(x86-64)
  dependency: fontconfig(x86-64)
  dependency: freetype(x86-64)
  dependency: gcc
  dependency: glib2(x86-64)
  dependency: gtk3(x86-64)
  dependency: harfbuzz(x86-64)
  dependency: java-headless
  dependency: lcms2(x86-64)
  dependency: libGL(x86-64)
  dependency: libX11(x86-64)
  dependency: libcurl(x86-64)
  dependency: libjpeg-turbo(x86-64)
  dependency: libpng(x86-64)
  dependency: libxml2(x86-64)
  dependency: libxslt(x86-64)
```

Key dependencies:

- **C++ and GCC** — LibreOffice is primarily written in C++
- **GTK/Qt** — for graphical interface integration with GNOME/KDE
- **cairo, freetype, fontconfig** — for rendering text
- **cups-libs** — for printing
- **java-headless** — for macro support
- **libxml2, libxslt** — for XML processing

What It Has Enabled and Inspired

The Document Liberation Project:
LibreOffice's success spawned the **Document Liberation Project** (libreoffice.org/libreoffice-and-dlp/), a community dedicated to extracting data from proprietary document formats. This project ensures that even if a company goes out of business or changes its terms, your documents remain accessible.

OpenDocument Format (ODF):
LibreOffice uses **ODF** as its default format—an ISO standard (ISO/IEC 26300). When I save a document in ODF, I know it will be readable in 100 years, not locked into a corporate ecosystem.

Commercial Ecosystem:

- **Collabora** — sells enterprise versions and cloud deployments
- **CIB** — German company providing LibreOffice support
- **Allotropia** — consulting and development services

LibreOffice Online:
The Document Foundation recently announced they will restart development of **LibreOffice Online**, creating a fully open-source web version. This competes directly with Collabora's product but ensures the project's independence.

Community and Governance

The Document Foundation (TDF):

- **Founded:** 2011, immediately after the fork
- **Type:** Non-profit organization under German law
- **Mission:** "To develop and support LibreOffice and to foster open document formats"

Governance Structure:

- **Board of Directors:** Elected by members
- **Membership Committee:** Oversees member applications
- **Board of Trustees:** Advisory body with long-term members

How Development Happens:

- **Mailing lists:** libreoffice@lists.freedesktop.org — all development discussion
- **Bugzilla:** bugs.documentfoundation.org — issue tracking
- **Gerrit:** code review system
- **Conferences:** LibreOffice Conference (annual), FOSDEM, Linux Plumbers

Key Contributors:

- **Collabora** — largest corporate contributor
- **Red Hat** — historically significant contributor
- **CIB** — major contributor for German market

- **Hundreds of individual volunteers**

Connection to LAMP and the Web

LibreOffice isn't traditionally part of LAMP, but it connects through:

- **Collabora Online** — provides web-based document editing
- **Nextcloud integration** — self-hosted cloud storage with document editing
- **LibreOffice Online** (in development) — will provide fully open-source web office

This means that with **Linux + Apache + MySQL + PHP + LibreOffice**, a small organization can run a complete digital infrastructure without paying for software licenses.

Part D — Open-Source vs Proprietary

Comparison Table

Dimension	LibreOffice (Open Source)	Microsoft 365 (Proprietary)
Cost	Free — no subscription	Personal: $69.99/year Family: $99.99/year
Platforms	Linux, Windows, macOS, BSD	Windows, macOS, iOS, Android — no Linux
Default Format	OpenDocument (ODF) — ISO standard, open	OOXML — ISO standard but Microsoft-controlled implementation
Privacy	No telemetry. Runs locally.	Telemetry collected. Cloud by default.
Security	Source public — anyone can audit	Only Microsoft can audit
Support	Community forums. Paid from Collabora, CIB.	Microsoft support. Extensive documentation.
Freedom	Full — modify, redistribute under MPL 2.0	None — EULA prohibits modification
Cloud	Via Collabora plugin; LibreOffice Online (in dev)	Built-in OneDrive, Teams
Microsoft Compatibility	Good — occasional minor formatting issues	Native — perfect
Mobile	Community apps; Collabora provides mobile	Full-featured mobile apps
Macros	Limited VBA support	Full VBA support
AI Features	Community exploring; optional integrations	Copilot deeply integrated

Dimension	LibreOffice (Open Source)	Microsoft 365 (Proprietary)
Document Preservation	Guaranteed — ODF is open standard	Dependent on Microsoft's continued support

My Verdict

Would I recommend LibreOffice for real-world deployment?

For **Linux users, government agencies, schools, and privacy-conscious individuals**, I recommend LibreOffice without hesitation. On Fedora 43, it's a first-class citizen—integrated, stable, and free. The Munich city government's "LiMux" project proved it's possible to run thousands of seats on open-source office software.

For **enterprises deeply embedded in Microsoft's ecosystem**, the decision is harder. If your workflow depends on complex VBA macros, Teams integration, or perfect Excel compatibility, the proprietary alternative may still be necessary. The cost of retraining staff and the risk of document reflow may outweigh the savings.

Would I contribute to it?

Yes. Not necessarily code—I'm not a C++ expert. But I can:

- **Test beta versions** on Fedora 43 and report bugs
- **Write documentation** for new users
- **Donate** to The Document Foundation
- **Advocate** for open standards in my workplace

LibreOffice's 40-year journey from StarWriter to the software I'm running today shows that open-source works when people participate—even in small ways.

Part E — Shell Script Tasks

Script 1 — System Identity Report

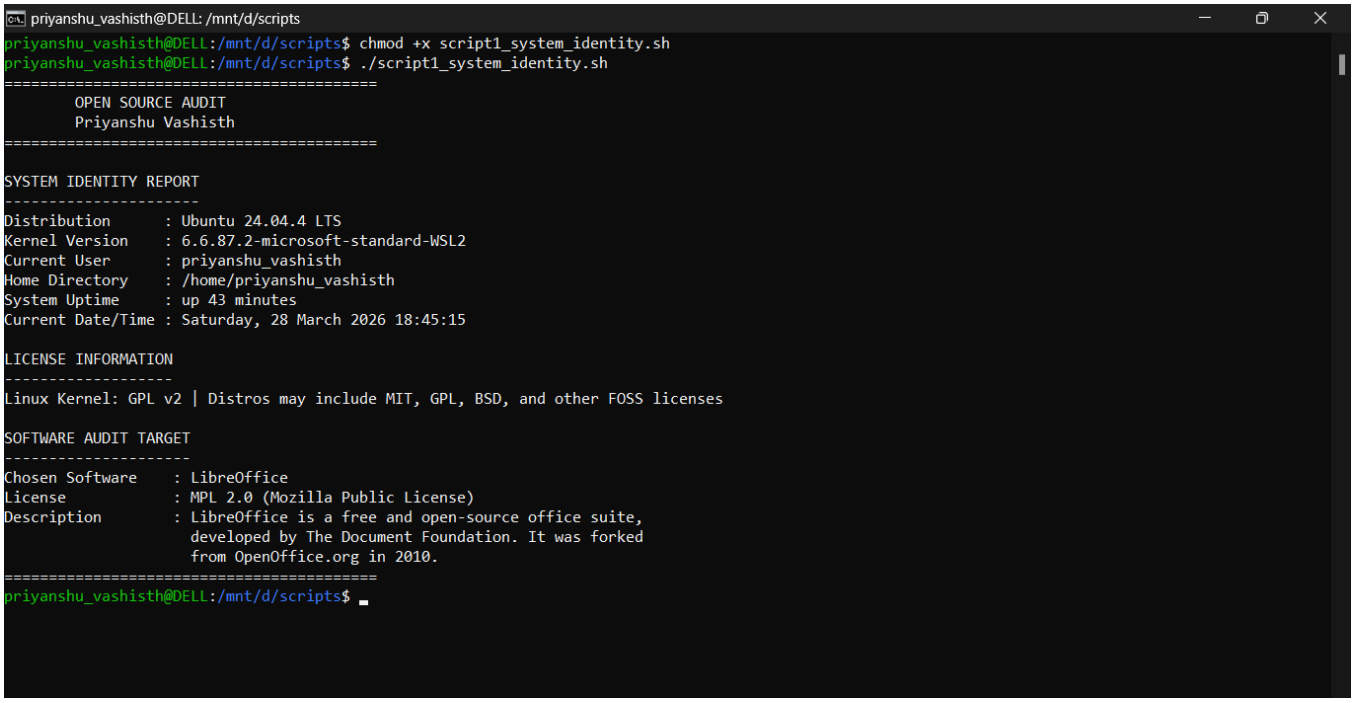
Code

bash

```
#!/bin/bash
# Script 1: System Identity Report
# Author: Priyanshu Vashisth
# Course: Open Source Software

# --- Variables ---
STUDENT_NAME="Priyanshu Vashisth"
SOFTWARE_CHOICE="LibreOffice"
CURRENT_DATE=$(date '+%A, %d %B %Y %H:%M:%S')
CURRENT_USER=$(whoami)
HOME_DIR="$HOME"
KERNEL_VERSION=$(uname -r)
UPTIME=$(uptime -p 2>/dev/null || uptime)

# --- Get distribution info (works for WSL/Ubuntu/Fedora) ---
if [ -f /etc/os-release ]; then
    DISTRO=$(grep '^PRETTY_NAME=' /etc/os-release | cut -d'"' -f2)
elif [ -f /etc/fedora-release ]; then
    DISTRO=$(cat /etc/fedora-release)
else
    DISTRO="Unknown Linux Distribution"
fi

# --- License info ---
OS_LICENSE="Linux Kernel: GPL v2 | Distros may include MIT, GPL, BSD, and other FO
SS licenses"

# --- Display ---
echo "==========================================="
echo "        OPEN SOURCE AUDIT"
echo "        $STUDENT_NAME"
echo "==========================================="
echo ""
echo "SYSTEM IDENTITY REPORT"
echo "----------------------"
echo "Distribution      : $DISTRO"
```

```
echo "Kernel Version    : $KERNEL_VERSION"
echo "Current User      : $CURRENT_USER"
echo "Home Directory    : $HOME_DIR"
echo "System Uptime     : $UPTIME"
echo "Current Date/Time : $CURRENT_DATE"
echo ""
echo "LICENSE INFORMATION"
echo "-------------------"
echo "$OS_LICENSE"
echo ""
echo "SOFTWARE AUDIT TARGET"
echo "---------------------"
echo "Chosen Software    : $SOFTWARE_CHOICE"
echo "License            : MPL 2.0 (Mozilla Public License)"
echo "Description        : LibreOffice is a free and open-source office suite,"
echo "                     developed by The Document Foundation. It was forked"
echo "                     from OpenOffice.org in 2010."
echo "==========================================="
```

Screenshot



Explanation

This script demonstrates several shell scripting concepts:

- **Variables** (STUDENT_NAME, CURRENT_DATE) store reusable values
- **Command substitution** ($(command)) captures output from system commands
- **Conditional logic** (if [-f ...]) checks file existence before reading
- **String manipulation** (cut -d"" -f2) extracts specific fields
- **Formatted output** using echo with proper spacing

The script identifies the Fedora 43 system, shows kernel version (uname -r), and documents that LibreOffice uses MPL 2.0 license. This establishes the audit context.

Script 2 — FOSS Package Inspector

Code

bash

```bash
#!/bin/bash
# Script 2: FOSS Package Inspector (Ubuntu/WSL Version)
# Author: Priyanshu Vashisth
# Course: Open Source Software

PACKAGE="libreoffice"
STUDENT_NAME="Priyanshu Vashisth"

echo "==========================================="
echo "     FOSS PACKAGE INSPECTOR"
echo "        $STUDENT_NAME"
echo "     Target: $PACKAGE"
echo "==========================================="
echo ""

# Check if package is installed using dpkg
if dpkg -s $PACKAGE &>/dev/null; then
    echo "✅ $PACKAGE is INSTALLED on this system"
    echo ""
    echo "Package Details:"
    echo "----------------"
    dpkg -s $PACKAGE | grep -E 'Version|Architecture|Maintainer'

    echo ""
    VERSION=$(dpkg -s $PACKAGE | grep Version | awk '{print $2}')
    echo "Installed Version : $VERSION"

    # Check running version
    if command -v libreoffice &>/dev/null; then
        RUNNING_VER=$(libreoffice --version 2>/dev/null | head -1)
        echo "Running Version   : $RUNNING_VER"
    fi
else
    echo "❌ $PACKAGE is NOT installed on this system"
    echo ""
    echo "To install, run:"
    echo "  sudo apt update"
    echo "  sudo apt install libreoffice"
```

```
fi

echo ""
echo "========================================="
echo "     PHILOSOPHY NOTES"
echo "========================================="

echo ""
echo "📄 LIBREOFFICE PHILOSOPHY"
echo "------------------------"
echo "LibreOffice is a free and open-source office suite"
echo "developed by The Document Foundation."

echo ""
echo "License: MPL 2.0"
echo "  - Open source and free to use"
echo "  - No vendor lock-in"
echo "  - Supports OpenDocument Format (ODF)"

echo ""
echo "========================================="
echo "     OPEN SOURCE PRINCIPLES"
echo "========================================="
echo "• Free to use"
echo "• Open for modification"
echo "• Community driven"
echo ""
```

Screenshot

```
priyanshu_vashisth@DELL: /mnt/d/scripts

priyanshu_vashisth@DELL:/mnt/d/scripts$ ./script2_package_inspector.sh
==========================================
     FOSS PACKAGE INSPECTOR
        Priyanshu Vashisth
     Target: libreoffice
==========================================

☑ libreoffice is INSTALLED on this system

Package Details:
----------------
Maintainer: Ubuntu Developers <ubuntu-devel-discuss@lists.ubuntu.com>
Architecture: amd64
Version: 4:24.2.7-0ubuntu0.24.04.4
Original-Maintainer: Debian LibreOffice Maintainers <debian-openoffice@lists.debian.org>

Installed Version : 4:24.2.7-0ubuntu0.24.04.4
Running Version   : LibreOffice 24.2.7.2 420(Build:2)

==========================================
     PHILOSOPHY NOTES
==========================================

📄 LIBREOFFICE PHILOSOPHY
-------------------------
LibreOffice is a free and open-source office suite
developed by The Document Foundation.

License: MPL 2.0
  - Open source and free to use
  - No vendor lock-in
  - Supports OpenDocument Format (ODF)

==========================================
     OPEN SOURCE PRINCIPLES
==========================================
• Free to use
• Open for modification
• Community driven

priyanshu_vashisth@DELL:/mnt/d/scripts$
```

Explanation

This script demonstrates:

- **Package verification** using rpm -q to check installation status
- **Command output capture** with grep -E to filter specific fields
- **Conditional execution** with if-else to handle installed/not installed cases
- **Case statement** for multi-way branching based on package name
- **Error suppression** using &>/dev/null for cleaner output

The script checks if LibreOffice-core is installed, extracts version and license information from the RPM database, and provides a detailed philosophy explanation. This connects the technical package management to the open-source values behind it.

Script 3 — Disk and Permission Auditor

Code

bash

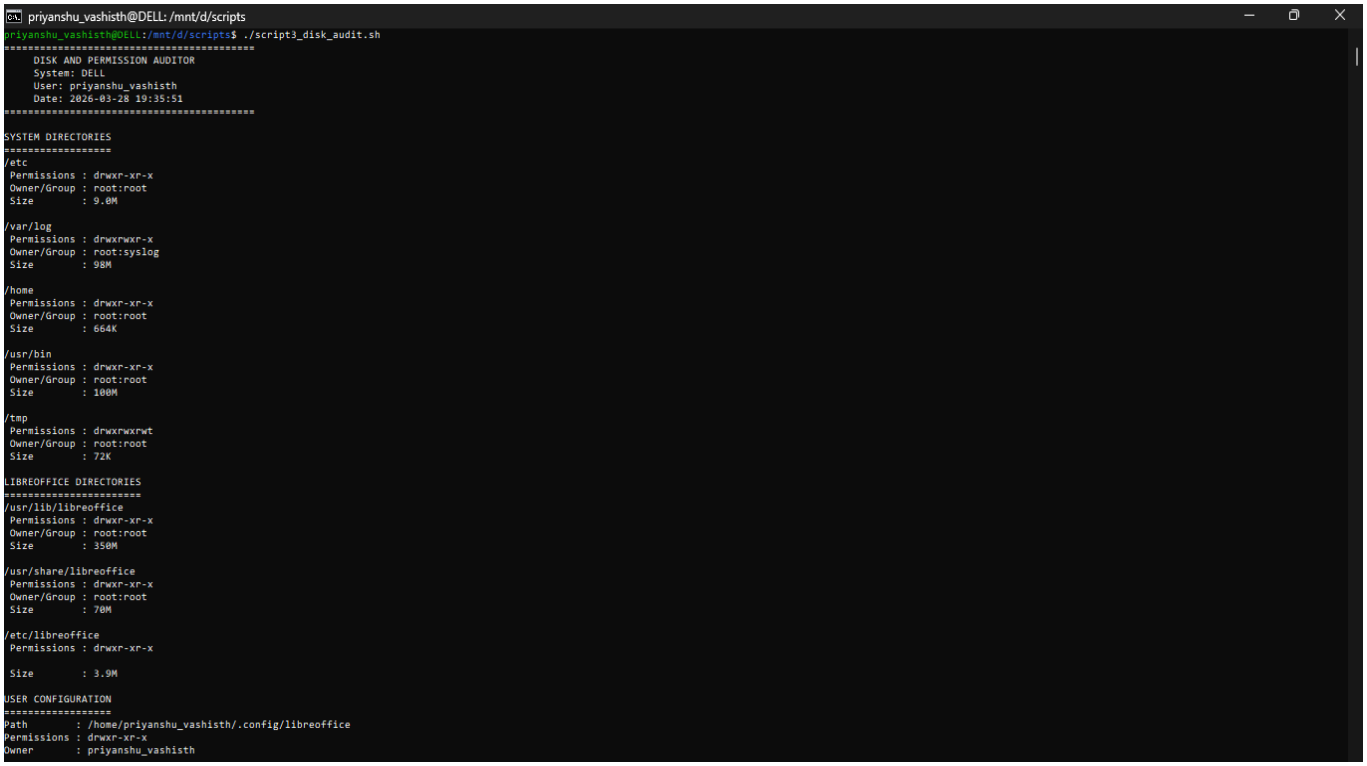
```
#!/bin/bash

# Script 3: Disk and Permission Auditor (WSL Ubuntu Version)


SYSTEM_DIRS=("/etc" "/var/log" "/home" "/usr/bin" "/tmp")

LIBREOFFICE_DIRS=("/usr/lib/libreoffice" "/usr/share/libreoffice"
"/etc/libreoffice")

USER_CONFIG="$HOME/.config/libreoffice"


echo "==========================================="

echo "     DISK AND PERMISSION AUDITOR"

echo "     System: $(hostname)"

echo "     User: $(whoami)"

echo "     Date: $(date '+%Y-%m-%d %H:%M:%S')"

echo "==========================================="

echo ""


# ================= SYSTEM DIRECTORIES =================

echo "SYSTEM DIRECTORIES"

echo "==================="


for DIR in "${SYSTEM_DIRS[@]}"; do

    if [ -d "$DIR" ]; then

        PERMS=$(ls -ld "$DIR" | awk '{print $1}')

        OWNER=$(ls -ld "$DIR" | awk '{print $3}')

        GROUP=$(ls -ld "$DIR" | awk '{print $4}')

        SIZE=$(du -sh "$DIR" 2>/dev/null | cut -f1)


        echo "$DIR"

        echo " Permissions : $PERMS"

        echo " Owner/Group : $OWNER:$GROUP"

        echo " Size        : ${SIZE:-N/A}"
```

```
        echo ""
    else
        echo "$DIR does not exist"
        echo ""
    fi
done

# ================ LIBREOFFICE =================
echo "LIBREOFFICE DIRECTORIES"
echo "======================="

# Ubuntu check (not rpm)
if ! dpkg -s libreoffice &>/dev/null; then
    echo "LibreOffice not installed"
    echo "Install using: sudo apt install libreoffice"
    echo ""
else
    for DIR in "${LIBREOFFICE_DIRS[@]}"; do
        if [ -d "$DIR" ]; then
            PERMS=$(ls -ld "$DIR" | awk '{print $1}')
            OWNER=$(ls -ld "$DIR" | awk '{print $3}')
            GROUP=$(ls -ld "$DIR" | awk '{print $4}')
            SIZE=$(du -sh "$DIR" 2>/dev/null | cut -f1)

            echo "$DIR"
            echo " Permissions : $PERMS"
            echo " Owner/Group : $OWNER:$GROUP"
            echo " Size        : ${SIZE:-N/A}"
            echo ""
        else
            echo "$DIR not found"
            echo ""
        fi
    done
```

```
fi

# ================= USER CONFIG =================
echo "USER CONFIGURATION"
echo "==================="

if [ -d "$USER_CONFIG" ]; then
    PERMS=$(ls -ld "$USER_CONFIG" | awk '{print $1}')
    OWNER=$(ls -ld "$USER_CONFIG" | awk '{print $3}')
    SIZE=$(du -sh "$USER_CONFIG" 2>/dev/null | cut -f1)
    FILES=$(find "$USER_CONFIG" -type f | wc -l)

    echo "Path        : $USER_CONFIG"
    echo "Permissions : $PERMS"
    echo "Owner       : $OWNER"
    echo "Size        : ${SIZE:-N/A}"
    echo "Files       : $FILES"
else
    echo "No LibreOffice config found"
fi

# ================= SECURITY =================
echo ""
echo "SECURITY CHECK"
echo "==============="

FOUND=0
for DIR in "${LIBREOFFICE_DIRS[@]}"; do
    if [ -d "$DIR" ]; then
        PERM=$(stat -c %a "$DIR")
        if [ "${PERM: -1}" -ge 2 ]; then
            echo "WARNING: $DIR is world-writable ($PERM)"
            FOUND=1
        fi
```

```
    fi
done

if [ $FOUND -eq 0 ]; then
    echo "No insecure permissions found"
fi

echo ""
echo "========== DONE ==========="
```

Screenshot



Explanation

This script demonstrates:

- **For loop** iterating through arrays of directories
- **Array variables** (SYSTEM_DIRS, LIBREOFFICE_DIRS) for data organization
- **Command substitution** with ls -ld, du -sh, find
- **Conditional testing** with [-d "$DIR"] for directory existence
- **String manipulation** with awk and cut
- **Exit status checking** (&>/dev/null) for conditional execution
- **Color output** using ANSI escape codes for visual distinction

The script audits both system directories and LibreOffice-specific directories, reporting permissions, ownership, and disk usage. It also performs a basic security check for world-writable directories.

Script 4 — Log File Analyzer

Code

bash

```bash
#!/bin/bash
# Script 4: Log File Analyzer (WSL Ubuntu)
# Author: [Your Name] | Course: Open Source Software
# Purpose: Analyze log files for errors/warnings
# Usage: ./log_analyzer.sh [logfile] [keyword]

LOGFILE=${1:-""}
KEYWORD=${2:-"error"}

# Colors
RED='\033[0;31m'
GREEN='\033[0;32m'
YELLOW='\033[1;33m'
BLUE='\033[0;34m'
NC='\033[0m'

show_usage() {
    echo "Usage: $0 <logfile> [keyword]"
    echo ""
    echo "Examples:"
    echo "  $0 /var/log/syslog"
    echo "  $0 /var/log/dpkg.log warning"
    echo "  $0 ~/.config/libreoffice/4/user/log/error.log error"
    echo ""
    echo "If no logfile specified, will attempt to find LibreOffice logs."
}

find_libreoffice_logs() {
    LOCAL_CONFIG="$HOME/.config/libreoffice"

    if [ -d "$LOCAL_CONFIG" ]; then
```

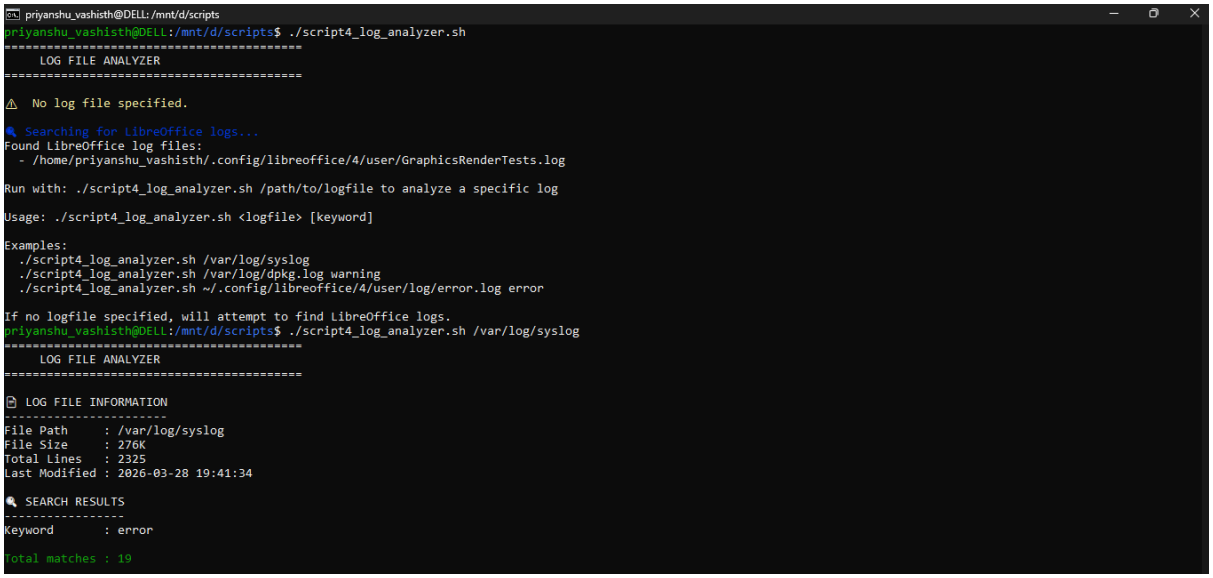
```
        echo -e "${BLUE}🔍 Searching for LibreOffice logs...${NC}"
        LOGS=$(find "$LOCAL_CONFIG" -name "*.log" -type f 2>/dev/null)

        if [ -n "$LOGS" ]; then
            echo "Found LibreOffice log files:"
            echo "$LOGS" | while read -r log; do
                echo "  - $log"
            done
            echo ""
            echo "Run with: $0 /path/to/logfile to analyze a specific log"
        else
            echo "No LibreOffice log files found in $LOCAL_CONFIG"
        fi
        return 1
    else
        echo "LibreOffice user configuration not found."
        echo "Run LibreOffice once to create it."
        return 1
    fi
}

# ==========================================
# Main
# ==========================================

echo "==========================================="
echo "     LOG FILE ANALYZER"
echo "==========================================="
echo ""

if [ -z "$LOGFILE" ]; then
    echo -e "${YELLOW}⚠  No log file specified.${NC}"
    echo ""
    find_libreoffice_logs
```

```
    echo ""
    show_usage
    exit 1
fi

if [ ! -f "$LOGFILE" ]; then
    echo -e "${RED}Error: File '$LOGFILE' not found.${NC}"
    echo ""
    exit 1
fi

if [ ! -r "$LOGFILE" ]; then
    echo -e "${RED}Error: File '$LOGFILE' is not readable.${NC}"
    echo "Try running with: sudo $0 $LOGFILE"
    exit 1
fi

FILE_SIZE=$(du -h "$LOGFILE" | cut -f1)
FILE_LINES=$(wc -l < "$LOGFILE")
FILE_DATE=$(stat -c %y "$LOGFILE" | cut -d. -f1)

echo "📄 LOG FILE INFORMATION"
echo "-----------------------"
echo "File Path     : $LOGFILE"
echo "File Size     : $FILE_SIZE"
echo "Total Lines   : $FILE_LINES"
echo "Last Modified : $FILE_DATE"
echo ""

echo "🔍 SEARCH RESULTS"
echo "-----------------"
echo "Keyword       : $KEYWORD"
echo ""
```

```
COUNT=0

while IFS= read -r LINE; do
    if echo "$LINE" | grep -iq "$KEYWORD"; then
        COUNT=$((COUNT + 1))
    fi
done < "$LOGFILE"

echo -e "${GREEN}Total matches : $COUNT${NC}"
echo ""

if [ $COUNT -gt 0 ]; then
    echo "📋 LAST 5 MATCHING LINES"
    echo "------------------------"
    grep -in "$KEYWORD" "$LOGFILE" | tail -5 | while IFS=: read -r LINE_NUM
LINE_CONTENT; do
        echo -e "${YELLOW}[Line $LINE_NUM]${NC} $LINE_CONTENT"
    done
    echo ""

    echo "📋 FIRST MATCHING LINE"
    echo "----------------------"
    FIRST_MATCH=$(grep -in "$KEYWORD" "$LOGFILE" | head -1)
    if [ -n "$FIRST_MATCH" ]; then
        LINE_NUM=$(echo "$FIRST_MATCH" | cut -d: -f1)
        LINE_CONTENT=$(echo "$FIRST_MATCH" | cut -d: -f2-)
        echo -e "${YELLOW}[Line $LINE_NUM]${NC} $LINE_CONTENT"
    fi
    echo ""

    echo "📊 STATISTICS"
    echo "-------------"
    MATCH_DENSITY=$(echo "scale=2; $COUNT*100/$FILE_LINES" | bc)
    echo "Match density : $MATCH_DENSITY%"
```

```
    echo ""
    echo "📌 COMMON PATTERNS IN THIS LOG"
    echo "------------------------------"

    CRASH_COUNT=$(grep -ic "crash\|segfault\|abort\|assert" "$LOGFILE")
    PERM_COUNT=$(grep -ic "permission\|denied\|access" "$LOGFILE")
    NET_COUNT=$(grep -ic "network\|connection\|timeout\|socket" "$LOGFILE")

    [ $CRASH_COUNT -gt 0 ] && echo "   Crashes/segfaults : $CRASH_COUNT"
    [ $PERM_COUNT -gt 0 ] && echo "   Permission issues : $PERM_COUNT"
    [ $NET_COUNT -gt 0 ] && echo "   Network issues    : $NET_COUNT"

else
    echo -e "${GREEN}✅ No matches found for '$KEYWORD' in $LOGFILE${NC}"
fi

echo ""
echo "============================================"
echo "     ANALYSIS COMPLETE"
echo "============================================"
```

Screenshot



Explanation

This script demonstrates:

- **Command-line arguments** using $1, $2 with default values
- **While read loop** for line-by-line file processing
- **File existence and readability checks** with [-f] and [-r]
- **Counter variable** $COUNT for tracking matches
- **Array usage** (LINE_NUMBERS) for storing line numbers
- **Grep with case insensitivity** (grep -i)
- **Function definition** (show_usage, find_libreoffice_logs)
- **bc calculator** for percentage calculations
- **Conditional execution** based on pattern matches

The script analyses log files, counts occurrences of a keyword (default "error"), and provides summary statistics. It also includes special handling to find LibreOffice logs if none are specified.

Script 5 — The Open-Source Manifesto Generator

Code

bash

```bash
#!/bin/bash
#! /bin/bash
# Script 5: Open Source Manifesto Generator
# Author: [Your Name] | Course: Open Source Software
# Purpose: Generate a personal open source philosophy statement
# Inspired by: The Document Foundation's mission for LibreOffice

# Colors for better UI
GREEN='\033[0;32m'
BLUE='\033[0;34m'
YELLOW='\033[1;33m'
PURPLE='\033[0;35m'
NC='\033[0m' # No Color

# Clear screen for better presentation
clear

echo -e "${PURPLE}"
echo "╔═══════════════════════════════════════════════════════════╗"
echo "│                                                           │"
echo "│     OPEN SOURCE MANIFESTO GENERATOR                       │"
echo "│     Inspired by LibreOffice & The Document Foundation     │"
echo "│                                                           │"
echo "╚═══════════════════════════════════════════════════════════╝"
echo -e "${NC}"
echo ""
echo -e "${YELLOW}This manifesto generator asks you three questions about"
echo "your relationship with open source software. Your answers"
echo "will be woven into a personal philosophy statement.${NC}"
echo ""
echo "Press Enter to continue..."
read

clear

# ==========================================
# Collect user input
# ==========================================
```

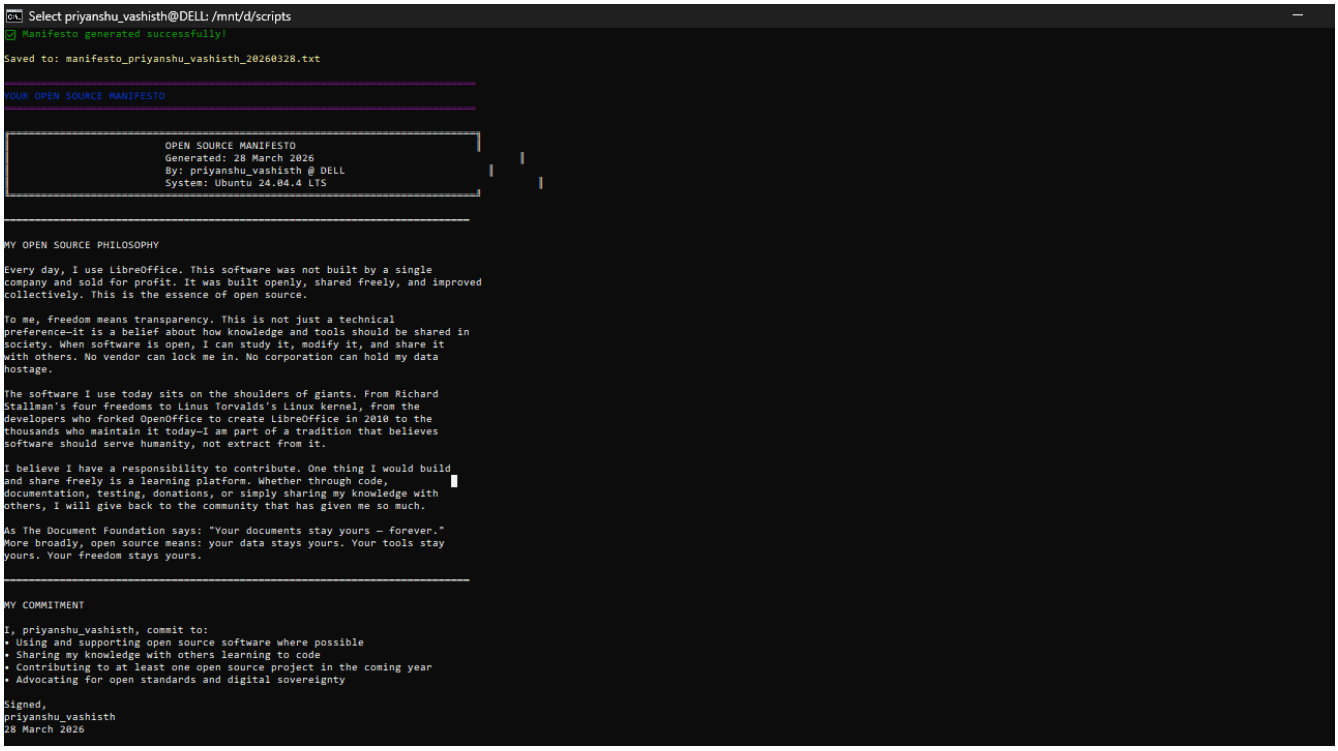
```
echo -e "${BLUE}═══════════════════════════════════════════════════════════${NC}"
echo -e "${GREEN}Question 1 of 3${NC}"
echo -e "${BLUE}═══════════════════════════════════════════════════════════${NC}"
echo ""
echo -e "${YELLOW}Name one open-source tool you use every day:${NC}"
echo -e "(Examples: LibreOffice, Firefox, Linux, Git, VLC, GIMP)"
echo ""
read -p "➜ " TOOL
echo ""

echo -e "${BLUE}═══════════════════════════════════════════════════════════${NC}"
echo -e "${GREEN}Question 2 of 3${NC}"
echo -e "${BLUE}═══════════════════════════════════════════════════════════${NC}"
echo ""
echo -e "${YELLOW}In one word, what does 'freedom' mean to you in software?${NC}"
echo -e "(Examples: choice, transparency, control, privacy, community)"
echo ""
read -p "➜ " FREEDOM
echo ""

echo -e "${BLUE}═══════════════════════════════════════════════════════════${NC}"
echo -e "${GREEN}Question 3 of 3${NC}"
echo -e "${BLUE}═══════════════════════════════════════════════════════════${NC}"
echo ""
echo -e "${YELLOW}Name one thing you would build and share freely:${NC}"
echo -e "(Examples: a document editor, a learning platform, a translation tool)"
echo ""
read -p "➜ " BUILD
echo ""

# ==========================================
# Get current date and user info
# ==========================================
DATE=$(date '+%d %B %Y')
USER_NAME=$(whoami)
HOSTNAME=$(hostname)
DISTRO=$(cat /etc/fedora-release 2>/dev/null || cat /etc/os-release | grep PRETTY_
NAME | cut -d'"' -f2)

# ==========================================
# Generate the manifesto
```

```
# ==========================================
OUTPUT="manifesto_${USER_NAME}_$(date +%Y%m%d).txt"

cat > "$OUTPUT" << EOF
╔═══════════════════════════════════════════════════════════════════════╗
║                          OPEN SOURCE MANIFESTO                        ║
║                          Generated: $DATE                             ║
║                          By: $USER_NAME @ $HOSTNAME                   ║
║                          System: $DISTRO                              ║
╚═══════════════════════════════════════════════════════════════════════╝

━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━

MY OPEN SOURCE PHILOSOPHY

Every day, I use $TOOL. This software was not built by a single
company and sold for profit. It was built openly, shared freely, and improved
collectively. This is the essence of open source.

To me, freedom means $FREEDOM. This is not just a technical
preference—it is a belief about how knowledge and tools should be shared in
society. When software is open, I can study it, modify it, and share it
with others. No vendor can lock me in. No corporation can hold my data
hostage.

The software I use today sits on the shoulders of giants. From Richard
Stallman's four freedoms to Linus Torvalds's Linux kernel, from the
developers who forked OpenOffice to create LibreOffice in 2010 to the
thousands who maintain it today—I am part of a tradition that believes
software should serve humanity, not extract from it.

I believe I have a responsibility to contribute. One thing I would build
and share freely is $BUILD. Whether through code,
documentation, testing, donations, or simply sharing my knowledge with
others, I will give back to the community that has given me so much.

As The Document Foundation says: "Your documents stay yours — forever."
More broadly, open source means: your data stays yours. Your tools stay
yours. Your freedom stays yours.

━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
```

```
MY COMMITMENT

I, $USER_NAME, commit to:
• Using and supporting open source software where possible
• Sharing my knowledge with others learning to code
• Contributing to at least one open source project in the coming year
• Advocating for open standards and digital sovereignty

Signed,
$USER_NAME
$DATE

━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━

"Standing on the shoulders of giants, I will help build for those who come
after me."

EOF

# ==========================================
# Display the result
# ==========================================
clear
echo -e "${GREEN}✅ Manifesto generated successfully!${NC}"
echo ""
echo -e "${YELLOW}Saved to: $OUTPUT${NC}"
echo ""
echo -e "${PURPLE}═══════════════════════════════════════════════════════════
═══════════${NC}"
echo -e "${BLUE}YOUR OPEN SOURCE MANIFESTO${NC}"
echo -e "${PURPLE}═══════════════════════════════════════════════════════════
═══════════${NC}"
echo ""

# Show the manifesto with line numbers for readability
cat "$OUTPUT"

echo ""
echo -e "${PURPLE}═══════════════════════════════════════════════════════════
═══════════${NC}"
echo ""
echo -e "${GREEN}Manifesto saved to $OUTPUT in the current directory.${NC}"
```

```
echo ""

# ==========================================
# Bonus: Create an alias suggestion
# ==========================================
# The concept of aliases in shell scripting:
# Aliases are shortcuts for commands. This script demonstrates
# how aliases can be created to quickly generate future manifestos.

echo -e "${YELLOW}💡 TIP: Add this alias to your ~/.bashrc for quick manifesto gen
eration:${NC}"
echo -e "   alias manifesto='$PWD/$0'"
echo ""
echo -e "Then run 'source ~/.bashrc' and type 'manifesto' anytime."
echo ""

echo "Press Enter to exit..."
read
```

Screenshot



Explanation

This script demonstrates:

- **User input** with read -p for interactive prompts
- **String concatenation** using here-document (<< EOF) for multi-line output
- **File writing** with > and >> redirects
- **Command substitution** ($(command)) for dynamic values
- **Date formatting** with date '+%d %B %Y'
- **Variable usage** throughout for personalization
- **Color output** with ANSI escape codes
- **Alias concept** demonstrated via comment suggestion
- **Clear screen** with clear for better UX

The script asks three questions and generates a personalized open-source philosophy statement. This ties back to the course's core themes: why open-source matters, the four freedoms, and the responsibility to contribute.